C

IB MATH: APPLICATIONS & INTERPRETATION • UNIT 3.01

Gradient Descent & Optimization

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Gradient Descent & Optimization Practice Exercises

1 MCQ

In gradient descent, a weight currently has value $w = 20$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 8$ and the learning rate is $\eta = 0.2$, find the updated weight w'.

A 12.00

B 21.60

C 18.40

D 19.80

2 MCQ

In gradient descent, a weight currently has value $w = 30$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 12$ and the learning rate is $\eta = 0.3$, find the updated weight w'.

A 18.00

B 33.60

C 26.40

D 29.70

3 MCQ

In gradient descent, a weight currently has value $w = 60$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 24$ and the learning rate is $\eta = 0.6$, find the updated weight w'.

A 36.00

B 74.40

C 45.60

D 59.40

4 MCQ

Consider the scalar loss function $L(w) = 2w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 2$.

A $\nabla L(2) = 16$

B $\nabla L(2) = 8$

C $\nabla L(2) = 0$

D $\nabla L(2) = -4$

5 MCQ

Consider the scalar loss function $L(w) = 3w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 4$.

A $\nabla L(4) = 16$

B $\nabla L(4) = 24$

C $\nabla L(4) = 32$

D $\nabla L(4) = 4$

6 MCQ

Consider the scalar loss function $L(w) = 4w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 6$.

A $\nabla L(6) = 40$

B $\nabla L(6) = 48$

C $\nabla L(6) = 56$

D $\nabla L(6) = 16$

7 MCQ

Consider the scalar loss function $L(w) = 5w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 8$.

A $\nabla L(8) = 72$

B $\nabla L(8) = 80$

C $\nabla L(8) = 88$

D $\nabla L(8) = 32$

8 MCQ

Consider the scalar loss function $L(w) = 6w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 10$.

A $\nabla L(10) = 112$

B $\nabla L(10) = 120$

C $\nabla L(10) = 128$

D $\nabla L(10) = 52$

9 MCQ

Consider the quadratic loss function $L(w) = w^2 - 4w + 3$. Find the value of w where the function reaches its minimum.

A $w = 3$

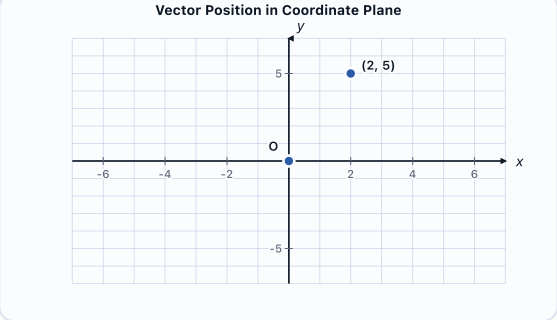
B $w = 2$

C $w = 4$

D $w = 1$

10 MCQ

Consider a univariate regression model where the predicted output $\hat{y}$ is given by $\hat{y} = ax + b$. The loss function is the Mean Squared Error (MSE), defined as $L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. For a single data point $(x_1, y_1) = (2, 5)$, the loss is $L = (5 - (2a + b))^2$. The gradient of the loss function with respect to a is $\frac{\partial L}{\partial a} = -2(5 - (2a + b)) \cdot 2$. If during gradient descent, with a learning rate of $\eta = 0.1$, the parameters a and b are updated from $a_{old} = 1$ and $b_{old} = 1$, to $a_{new} = 0.8$ and $b_{new} = 0.8$, what is the value of b that would have resulted in $\frac{\partial L}{\partial b} = 0$ for this single data point, assuming $a = 1$ and $b = 1$ were the values *before* the gradient calculation that led to the update to $a_{new} = 0.8, b_{new} = 0.8$?



A 3
B 4
C 6
D 5

11 MCQ

Considering gradient descent, a weight currently has value $w = 30$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 12$ and the learning rate is $\eta = 0.3$, find the updated weight w'.

A 18.00
B 33.60
C 29.70
D 26.40

12 MCQ

Examining gradient descent, a weight currently has value $w = 60$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 24$ and the learning rate is $\eta = 0.6$, find the updated weight w'.

A 59.40
B 36.00
C 45.60
D 74.40

13 MCQ

Consider the following quadratic loss function for a regression problem: $L(y, \hat{y}) = (y - \hat{y})^2$. If the predicted value $\hat{y} = 5$ and the true value $y = 3$, what is the value of the loss function?

A -2
B 9
C 4
D 2

14 MCQ

An Ordinary Least Squares linear model $\hat{f}(x) = 23x + 14$ is evaluated at $x = 19$. Determine the predicted response $\hat{f}(19)$.

A 451

B 455

C 474

D 437

15 MCQ

Consider the following quadratic loss function: $L(w) = 3w^2 - 12w + 10$. What is the value of w that minimizes this loss function?

A 2

B 12

C 1

D 4

16 MCQ

Consider the scalar loss function $L(w) = 7w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 12$.

A $\nabla L(12) = 168$

B $\nabla L(12) = 176$

C $\nabla L(12) = 160$

D $\nabla L(12) = 76$

17 MCQ

A 2D loss surface is defined as $L(w_1, w_2) = 2w_1^2 + 3w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

A $\nabla L = [10, -8]^T$

B $\nabla L = [8, 6]^T$

C $\nabla L = [8, -6]^T$

D $\nabla L = [4, -3]^T$

18 MCQ

A 2D loss surface is defined as $L(w_1, w_2) = 3w_1^2 + 5w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

A $\nabla L = [14, -12]^T$

B $\nabla L = [12, 10]^T$

C $\nabla L = [6, -5]^T$

D $\nabla L = [12, -10]^T$

19 MCQ

In an applied problem, a 2D loss surface is defined as $L(w_1, w_2) = 4w_1^2 + 7w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

A $\nabla L = [18, -16]^T$

B $\nabla L = [16, -14]^T$

C $\nabla L = [16, 14]^T$

D $\nabla L = [8, -7]^T$

20 MCQ

Suppose a 2D loss surface is defined as $L(w_1, w_2) = 5w_1^2 + 9w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

A $\nabla L = [20, -18]^T$

B $\nabla L = [22, -20]^T$

C $\nabla L = [20, 18]^T$

D $\nabla L = [10, -9]^T$

21 MCQ

An observed 2D loss surface is defined as $L(w_1, w_2) = 6w_1^2 + 11w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

A $\nabla L = [26, -24]^T$

B $\nabla L = [24, -22]^T$

C $\nabla L = [24, 22]^T$

D $\nabla L = [12, -11]^T$

22 MCQ

Consider a 2D loss surface is defined as $L(w_1, w_2) = 7w_1^2 + 13w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

A $\nabla L = [30, -28]^T$

B $\nabla L = [28, 26]^T$

C $\nabla L = [28, -26]^T$

D $\nabla L = [14, -13]^T$

23 MCQ

Consider minimizing the convex loss $L(w) = (w + 3)^2$ (Run #1) using gradient descent with learning rate $\eta = 0.25$ starting at $w_0 = -1$. Compute $w_1 = w_0 - \eta L'(w_0)$.

A -3.50

B 0.00

C -2

D 4

24 MCQ

Consider minimizing the convex loss $L(w) = 2(w + 2)^2$ (Run #2) using gradient descent with learning rate $\eta = 0.25$ starting at $w_0 = 1$. Compute $w_1 = w_0 - \eta L'(w_0)$.

A -2

B -3.50

C 4.00

D 12

25 MCQ

Consider minimizing the convex loss $L(w) = 3(w + 1)^2$ (Run #3) using gradient descent with learning rate $\eta = 0.25$ starting at $w_0 = 3$. Compute $w_1 = w_0 - \eta L'(w_0)$.

A -3

B 9.00

C 24

D -4.50